

Structural model robustness checks in PLS-SEM

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Abstract

Partial least squares structural equation modeling (PLS-SEM) has become a standard tool for analyzing complex inter-relationships between observed and latent variables in tourism and numerous other fields of scientific inquiry. Along with the recent surge in the method's use, research has contributed several complementary methods for assessing the robustness of PLS-SEM results. Although these improvements are documented in extant literature, research on tourism has been slow to adopt the relevant complementary methods. This article illustrates the use of recent advances in PLS-SEM, designed to ensure structural model results' robustness in terms of nonlinear effects, endogeneity, and unobserved heterogeneity in a PLS-SEM framework. Our overarching aim is to encourage the routine use of these complementary methods to increase methodological rigor in the field.

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Keywords

endogeneity, FIMIX-PLS, nonlinear, partial least squares, PLS-SEM, robustness checks, structural equation modeling

Introduction

Partial least squares structural equation modeling (PLS-SEM) has become a standard tool for analyzing complex inter-relationships between observed and latent variables in social science research, as shown by numerous studies that have reviewed its use in different fields, such as hospitality management, marketing, and tourism (Table 1). PLS-SEM’s impact extends into various other fields of scientific inquiry such as engineering (e.g. Durdyev et al., 2018), medicine (e.g. Menni et al., 2018), and psychology (e.g. Willaby et al., 2015). PLS-SEM has various advantageous features that have contributed to its increasing dissemination during the last decade. These features include the technique’s ability to handle complex models with relatively little data (Hair et al., 2017c), to estimate formatively specified measurement models (Sarstedt et al., 2016a), and produce determinate latent variable scores (Rigdon et al., 2019). Published research has regularly emphasized that PLS-SEM is particularly appealing for applied science by allowing the testing of hypothesized relationships while taking a prediction focus in the model estimation (Cepeda et al., 2016; Evermann and Tate, 2016; Sarstedt et al., 2017a). PLS-SEM thus overcomes the apparent dichotomy between *explanation*, which academic research usually emphasizes, and *prediction*, which is required to derive managerial implications (Hair et al., 2019; Shmueli et al., 2016, 2018). In addition, the many controversial debates about the method’s merits and limitations, witnessed in different research fields (Khan et al., 2019), have increased awareness of it (Petter, 2018). As part of these debates, researchers identified blind spots in PLS-SEM, which methodologists were quick to fill by developing methodological improvements (Franke and Sarstedt, 2019; Henseler et al., 2015; Sharma et al., 2019b).

Table 1. Studies that review the use of PLS-SEM in business disciplines.

Business discipline	Reference	Period	Number of studies
Marketing	Hair et al. (2012b)	1981–2010	204
Strategic management	Hair et al. (2012a)	1981–2010	37
Management information systems	Ringle et al. (2012)	1992–2001	65
Management information systems	Hair et al. (2017a)	2010–2014	57
International business	Richter et al. (2016)	1990–2013	45
Human resource management	Ringle et al. (2019)	1985–2014	114
Operations management	Peng and Lai (2012)	2000–2011	42
Supply chain management	Kaufmann and Gaeckler (2015)	2002–2013	75
Accounting	Lee et al. (2011)	2005–2011	20
Accounting	Nitzl (2016)	1980–2013	37
Tourism	do Valle and Assaker (2016)	2000–2014	44
Hospitality	Ali et al. (2018b)	2001–2015	29
Hospitality and tourism	Usakli and Kucukergin (2018)	2000–2017	206

PLS-SEM: partial least squares structural equation modeling.

A particularly prominent research stream in PLS-SEM has provided complementary methods for assessing the results' robustness (Hair et al., 2018b; Latan, 2018). Such robustness checks have become common in regression-based studies, where researchers examine "how certain 'core' regression coefficient estimates behave when the regression specification is modified in some way, typically by adding or removing regressors." (Lu and White, 2014: 194). However, despite being a regression-based method, PLS-SEM researchers rarely engage in such model modifications and comparisons (Sharma et al. 2019a). Similarly, applications of specification checks (Callaghan et al., 2007) and specification searches, for example using ant colony optimization (Marcoulides and Drezner, 2003), have not flourished in PLS-SEM. Nevertheless, there have been some efforts in PLS-SEM-based research to test the robustness of structural model parameters by extending the model or running complimentary methods. For example, researchers have stressed the importance of considering potential nonlinearities in the model relationships, which often map relationships between unobservable phenomena better when they do occur. In their study on residents' support for continued tourism development activities, Rasoolimanesh et al. (2018: 1) note that if a relationship is "erroneously assumed to be linear then not only will the true relationship be underestimated, but the effects of this relationship might register as weak or non-significant." Similarly, researchers have proposed approaches for identifying and treating unobserved heterogeneity and endogeneity that—if not taken into consideration—can pose serious threats to the results' validity (Becker et al., 2013; Hult et al., 2018). Summarizing the role of structural robustness checks, Latan (2018: 68) recently noted that "these issues must be urgently addressed, and reported in the results of PLS-SEM."

Although these advances are documented in extant literature (Hair et al., 2018b; Latan and Noonan, 2017), authors, reviewers, and editors may find keeping up with the latest developments daunting, leading to the relevant complementary methods only being adopted slowly. With this issue in mind, this article explains the nature and illustrates the use of recent PLS-SEM improvements, designed to ensure the results' robustness. Our focus is on assessing the structural model estimates' robustness, which is of fundamental concern when testing theoretical frameworks and deriving managerial recommendations. We start by first discussing problems related to and approaches for handling nonlinear effects, endogeneity, and unobserved heterogeneity in a PLS-SEM framework. We subsequently illustrate the use of these approaches on a tourism marketing example. Finally, we discuss further research avenues related to model robustness checks in PLS-SEM.

Robustness checks in PLS-SEM

Nonlinear effects

When estimating path models, researchers usually assume that the relationships between the constructs are linear by nature. While linear relationships generally approximate relations found in reality well, this is not necessarily always the case (Ahrholdt et al., 2019). Nevertheless, studies using PLS-SEM have only rarely considered nonlinear effects (Table 2). Specifically, our review of all studies using PLS-SEM in all tourism and hospitality journals indexed by Clarivate Analytics in its 2018 Journal Citations Reports in the "Hospitality, Leisure, Sport & Tourism" category, shows that only 9 of 375 studies (2.4%) estimated nonlinear effects, while 3 other papers mentioned nonlinearity as a limitation and/or future research direction. Those studies addressing nonlinearity do so by analyzing the nonlinear relationships between tourism resources and tourism

Table 2. Empirical studies employing or calling for the use of methods for the robustness assessment of PLS-SEM results.

Robustness method/ empirical study	Publication title	Tourism topic	Robustness assessment	Observations
Nonlinear effects				
Šerić et al. (2018)	<i>Current Issues in Tourism</i>	Customer-based brand equity in the hotel industry	Yes	Impact-asymmetry analysis as the second stage after PLS-SEM
Ali et al. (2018a)	<i>International Journal of Contemporary Hospitality Management</i>	Customer experience in the hotel industry; green practices in coffee shops; tourists' satisfaction with destination	No	Nonlinear effects analysis suggested for future research
Beldona et al. (2018)	<i>International Journal of Contemporary Hospitality Management</i>	Technological offerings in the hotel industry	No	Nonlinearity issues mentioned as study limitations
Wong and Lai (2018)	<i>International Journal of Hospitality Management</i>	Value co-creation activities at exhibitions	Yes	Impact-asymmetry analysis as the second stage after PLS-SEM
Rasoolimanesh et al. (2018)	<i>Journal of Destination Marketing & Management</i>	Residents' support for tourism destination development	Yes	Structural model assessments using nonlinear algorithms
Sarstedt et al. (2014)	<i>Journal of Leisure Research</i>	Soccer fan satisfaction	No	Nonlinear effects analysis suggested for future research
Camarero et al. (2019)	<i>Journal of Travel & Tourism Marketing</i>	Operating and funding performance of museums	Yes	Quadratic effect assessed by establishing interaction terms
Antón et al. (2018)	<i>Journal of Travel Research</i>	Tourists' experience value	Yes	Quadratic effect assessed by establishing interaction terms
Mazanec and Ring (2011)	<i>Tourism Economics</i>	Competitiveness in tourism destinations	Yes	Quadratic effect assessed by establishing interaction terms
Hew et al. (2018)	<i>Tourism Management</i>	Mobile social tourism shopping	Yes	Artificial neural network analysis as the second stage after PLS-SEM

(continued)

Table 2. (continued)

Robustness method/ empirical study	Publication title	Tourism topic	Robustness assessment	Observations
Mohsin et al. (2015)	<i>Tourism Management</i>	Staff turnover in the hotel industry	Yes	Quadratic effect assessed by establishing interaction terms
Yadegaridehkordi et al. (2018)	<i>Tourism Management</i>	Critical success factors in the hotel industry	Yes	Adaptive neuro-fuzzy inference systems modeling as the second stage after PLS-SEM
Endogeneity Marder et al. (2019)	<i>Journal of Travel Research</i>	Vicarious travel consumption	Yes	The authors ensure they avoid the risk of endogeneity prior to data collection by selecting the two most similar stimuli; they further conduct a paired-sample t-test to account for differences across the two stimuli; to eliminate the risk of endogeneity in future studies, researchers are advised to uncover emotions by measuring actual behavior and the psychological response in experiments
Unobserved heterogeneity Marques and Reis (2015)	<i>Annals of Tourism Research</i>	Preference for outdoor activities	Yes	The authors emphasize the importance of uncovering unobserved heterogeneity in the context of PLS-SEM and address this issue through the use of FIMIX-PLS; their results point to a four-class partition of the data and reveal that the relationships between environmental attitudes and preference for outdoor activities differ across heterogeneous classes
Assaker et al. (2020)	<i>Current Issues in Tourism</i>	Human capital and restaurant operators performance	Yes	The authors use REBUS-PLS to account for unobserved heterogeneity in their data; the results indicate a three-class partition of the units and reveal that the strength of the relationships between human capital factors and restaurant operators performance varies across the three segments
Sarstedt et al. (2014)	<i>Journal of Leisure Research</i>	Soccer fan satisfaction	No	The authors emphasize the importance of the assessment of unobserved heterogeneity in future research aiming to predict fan attendance using the proposed satisfaction index (FANSAT)

(continued)

Table 2. (continued)

Robustness method/ empirical study	Publication title	Tourism topic	Robustness assessment	Observations
So et al. (2018a)	<i>Journal of Travel Research</i>	Consumer brand identification	No	The authors propose that future studies replicating the model also address the issue of unobserved heterogeneity
Mazanec and Ring (2011)	<i>Tourism Economics</i>	Competitiveness in tourism destinations	Yes	The authors employ the FIMIX-PLS technique to account for unobserved heterogeneity in the data set of the destination competitiveness of 129 countries; the results indicate three segments of economies that differ in terms of how the constituents of competitiveness influence tourism performance
Amaro and Duarte (2015)	<i>Tourism Management</i>	Online purchases of travels	No	To better address consumers' needs and understand the factors that influence their intentions to purchase travels online, the authors propose that future studies employ the FIMIX-PLS technique to uncover different consumer segments
Rasoolimanesh et al. (2017)	<i>Tourism Management</i>	Tourism development	No	The authors speculate that the nonsignificant effect of urban residents' negative perceptions of tourism development on their community participation is due to heterogeneity; to further investigate this finding, the authors suggest that future studies should address the issue of unobserved heterogeneity
So et al. (2018b)	<i>Tourism Management</i>	Adoption of peer-to-peer accommodation	No	The authors contend that future studies replicating the model should also address the issue of unobserved heterogeneity, thereby enhancing the possibilities of refining the product offerings and management practices in accordance with the characteristics of different subgroups of consumers

Note: PLS-SEM; partial least squares structural equation modeling; FIMIX-PLS: finite mixture partial least squares.

competitiveness (Mazanec and Ring, 2011), hotel staff turnover and its antecedents (Mohsin et al., 2015), hotel customer-based brand equity and its drivers and consequences (Šerić et al., 2018), exhibitors' value co-creation activities and their satisfaction with exhibitions (Wong and Lai, 2018), residents' perceptions of tourism destination development effects and their support for this development (Rasoolimanesh et al., 2018), the operating/funding performance of museums and their creativity orientation (Camarero et al., 2019), perceived tourism experience value and tourists' experience intensity (Antón et al., 2018), mobile social tourism shopping and environmental stimuli (Hew et al., 2018), and hotel success and its critical factors (Yadegaridehkordi et al., 2018). Most of these studies have analyzed the nonlinear effects by means of interaction terms in their PLS path models (4 papers) or by using various techniques (impact-asymmetry analysis, artificial neural networks, and adaptive neuro-fuzzy inference systems) as second stage analyses (5 papers).

When the relationship between two constructs is nonlinear, the size of the effect between two constructs not only depends on the magnitude of the change in the exogenous construct but also on its value (Hair et al., 2018b). To test whether or not relationships are nonlinear, researchers can run Ramsey's (1969) regression equation specification error test (RESET). To implement the test in a PLS-SEM context, researchers first need to estimate the model and use the resulting construct scores as input for the RESET, which standard software packages, such as Stata (Mooi et al., 2018) and SPSS (for an implementation of the RESET in SPSS, see Sarstedt and Mooi, 2019), execute. The test considers quadratic effects, which are the most common, and cubic effects by default.

As this implementation of the RESET relies on construct scores estimated from a linear effects-only model, researchers should also check whether the specification of a nonlinear effect yields a significant result. To consider, for example, a quadratic effect in PLS-SEM (for more details and applications, see Hair et al., 2018a), researchers need to consider a polynomial model and add a quadratic term. This quadratic term is similar to an interaction term, which comprises the exogenous construct's interaction with itself (Rigdon et al., 2010). If the interaction term's effect is significant and positive (negative), the strength of the exogenous construct's effect increases (decreases) in the exogenous construct's higher values. Conversely, a nonsignificant interaction term offers evidence of the linear effect's robustness.

Endogeneity

The consideration of endogeneity is a key issue when applying regression-based methods such as PLS-SEM (Hult et al., 2018). Endogeneity in PLS-SEM occurs when a predictor construct is correlated with the error term of the dependent construct to which it is related (Bascle, 2008). This implies that the predictor construct not only explains the dependent construct but also its error term. While endogeneity can have several roots (Ebbes et al., 2016; Papies et al., 2016), it mostly arises from omitted constructs that correlate with one or more predictor constructs and with the dependent construct in a partial regression of the PLS path model.

In tourism applications, endogeneity may arise when, for example, researching the effect of visitor satisfaction on spending (Mortazavi, 2018), which requires considering all aspects of visitor satisfaction, such as their satisfaction with the price, venue, personnel, service quality, and many other aspects. However, considering all these, or at least the most salient aspects, in a single study is very challenging (Mortazavi, 2018). If an omitted construct affects both the dependent variable and one or more independent variable(s) in the path model, this will induce a correlation between the independent variable(s) and the error term (Wooldridge, 2013). Variation in the omitted

construct or variable will therefore manifest in the error term, thereby violating a key causal modeling assumption of regression analysis—the error term is assumed to have an expected value of zero (Wooldridge, 2013). Consequently, the coefficient estimates from standard regression analyses are biased, thereby becoming causally uninterpretable and potentially triggering type I and type II errors (Papies et al., 2016). Since PLS-SEM relies on regression analysis to estimate partial model structures, considering endogeneity is particularly relevant for studies that focus on explanation. However, our review of studies (Table 2) shows that tourism researchers do not consider endogeneity in their PLS-SEM analyses, despite a clear focus on explanation. Of a total of 375 papers employing the PLS-SEM approach published in the chosen tourism and hospitality journals, only Marder et al. (2019) make explicit reference to endogeneity. However, these authors do not follow an approach specific to regression-based methods such as PLS-SEM but address the endogeneity issue in the research design stage.

This tendency to overlook the endogeneity issue is not surprising given that methodological researchers have only recently started discussing this topic in the context of PLS-SEM (Benitez et al., 2016; Hult et al., 2018). Hult et al. (2018) reviewed the approaches used in a regression context and developed a systematic procedure for identifying and treating endogeneity in PLS-SEM. Their guideline builds on Park and Gupta's (2012) Gaussian copula approach, which controls for endogeneity by directly modeling the correlation between the endogenous variable and the error term by means of a copula. If the approach indicates the presence of an endogeneity issue, researchers should implement instrumental variables that are highly correlated with the independent constructs but are uncorrelated with the dependent construct's error term, to explain the sources of endogeneity (Bascle, 2008).

Unobserved heterogeneity

Latent class techniques for identifying and treating unobserved heterogeneity have a long-standing tradition in PLS-SEM (Sarstedt, 2008). Unobserved heterogeneity occurs when subgroups of data exist, which entail substantially different model estimates. In such a case, estimating the model based on the entire data set is bound to produce misleading results; for example, the positive and negative effects in each of the groups could cancel each other. Latent class techniques allow for identifying such subgroups of data and estimating group-specific parameters. Since the introduction of the initial approach, finite mixture PLS (FIMIX-PLS; Hahn et al., 2002), research on latent class analysis in PLS-SEM has made great progress, producing several techniques that differ in their scope and areas of application. Based on their review of the available approaches, Sarstedt et al. (2017b) have proposed a systematic procedure for identifying and treating unobserved heterogeneity that draws on a combination of latent class procedures. In the first step, researchers should run FIMIX-PLS to ascertain whether heterogeneity is an issue (or not). FIMIX-PLS is particularly valuable in this regard by producing model selection criteria that guide the decision on how many segments to retain from the data (Hair et al., 2016; Matthews et al., 2016). Sarstedt et al. (2011) evaluated the efficacy of various model selection criteria, demonstrating that researchers should jointly consider modified Akaike's information criterion with factor 3 (AIC₃) (Bozdogan, 1994) and consistent AIC (CAIC; Bozdogan, 1987). Additional considerations when deciding on the number of segments include the segments' fuzziness, as expressed by high values in the entropy statistic (EN) (Ramaswamy et al., 1993), and the segment sizes, which need to meet minimum sample size requirements (Kock and Hadaya, 2018).

If the metrics point to a one-segment solution or produce divergent results, researchers can conclude that unobserved heterogeneity does not significantly affect the data (Sarstedt et al., 2017b). This result offers evidence of the results' robustness. If, however, the analysis indicates a significant degree of heterogeneity, researchers should proceed by running the PLS prediction-oriented segmentation (PLS-POS; Becker et al., 2013) approach on the data to uncover the segment structure and estimate the segment-specific parameters. Based on these results, the researcher may aim at explaining the uncovered heterogeneity (e.g. age, income, sex, etc.). This can be done by means of contingency table analyses and an exhaustive Chi-squared automatic interaction detector (CHAID) analysis (Ringle et al., 2010), which allow for identifying explanatory variables that match the PLS-POS-based partition (Sarstedt et al., 2017b). A subsequent multigroup analysis (Chin and Dibbern, 2010; Hair et al., 2018b), combined with a measurement invariance assessment (Henseler et al., 2016), allows testing for significant differences in the group-specific parameter estimates.

While latent class techniques have frequently been used in tourism research in a variety of methodological contexts (e.g. Alegre et al., 2011; Fuentes-Blasco et al., 2017; López-Sánchez and Pulido-Fernández, 2016), their application in the context of PLS-SEM is scarce (Table 2). For example, Marques and Reis (2015) acknowledged the implications that the assumption of data homogeneity may have for model estimates and applied the FIMIX-PLS technique to address the issue of unobserved heterogeneity in the data set. Their findings demonstrate that the overall sample results of the relationships between environmental attitudes and the target construct of preference for outdoor activities are not representative of the four underlying segments of consumers that the FIMIX-PLS technique suggests. Nevertheless, the authors did not identify explanatory variables that match the four-segment partition produced by FIMIX-PLS (Sarstedt et al., 2016b). In another study, Mazanec and Ring (2011) used FIMIX-PLS to identify three distinct segments of economies that differ with regard to how the various competitiveness constituents translate into tourism performance. More recently, Assaker et al. (2020) used response-based procedure for detecting unit segments (REBUS-PLS), a distance measure-based method, to uncover unobserved heterogeneity in their data. Their segmentation results revealed that the strength of the relationships between human capital factors and restaurant operators' performance varies across the three-segment solution provided by REBUS-PLS.¹ Using the CHAID technique, Assaker et al. (2020) profile the three segments of restaurant operators by using a combination of explanatory variables (e.g. business characteristic variables, environmental hostility). Other authors in the tourism field (e.g. Chandel and Bansal, 2014; Rasoolimanesh et al., 2017; So et al., 2018a, 2018b) only note that future research should extend their analyses by running, for example, FIMIX-PLS on the data. For example, Amaro and Duarte (2015: 75) called for using FIMIX-PLS to better understand the factors that influence intentions to purchase travels online in different segments, thereby enhancing "the possibilities of developing online marketing strategies that meet the needs of each segment."

Empirical tourism research example

Model and data

To illustrate the structural model robustness checks, we draw on Amaro and Duarte's (2015) model on consumers' intentions to purchase travels online. The model's goal is to explain the effects of trust (*TRUST*) and communicability (*COMM*) on attitude (*ATT*) and, finally, the intention to purchase travels online (*IPTO*). Figure 1 illustrates the path model used in our analyses. All

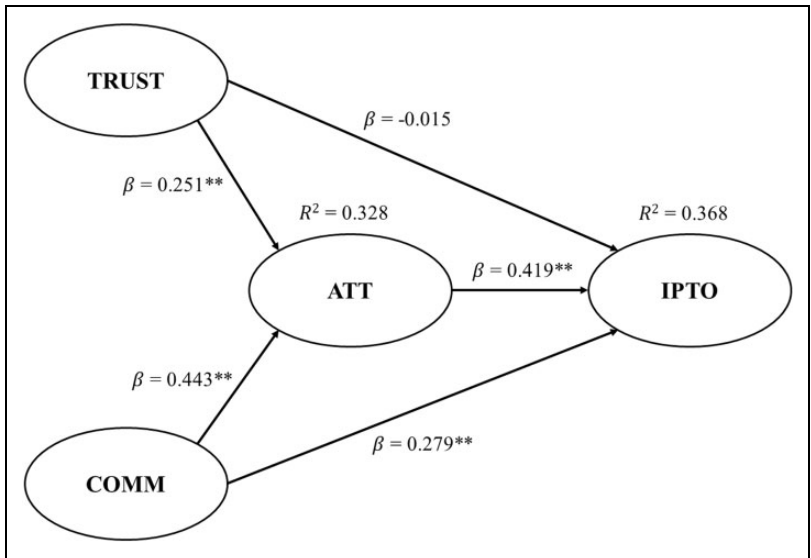


Figure 1. Consumers' intentions to purchase travels online.

Note. * p Value < 0.05; ** p value < 0.01.

constructs draw on reflective measurement models whose items are measured on seven-point Likert-type scales. Table 3 offers an overview of all the constructs and their measurement items.

The online survey-based data collection resulted in 500 responses from online trip purchasers who regularly visit popular travel websites (e.g. Booking, TripAdvisor, Expedia, Hotels, Travelocity, and AirBnB). The data set consists primarily of female Malay journey purchasers between 21 and 30 years with a monthly income of RM 2001 to RM 4000 (about 420 to 840 Euros). Table 4 offers an overview of the sample demographics.²

Our analysis uses the SmartPLS 3 software to estimate the model parameters (Ringle et al., 2015). In line with Hair et al.'s (2017b) recommendations, we use path weighting, with a maximum of 300 iterations, and a stop criterion of 10^{-7} ($=1.0E-07$) in the PLS-SEM algorithm settings.

Standard model assessment

The evaluation of the measurement models by means of standard evaluation criteria (Hair et al., 2017b; Ramayah et al., 2016; Sarstedt et al., 2017a) supports the measures' reliability and validity (Table 3). This also holds for discriminant validity assessment using Henseler et al.'s (2015) recently proposed heterotrait-monotrait (HTMT) criterion (Table 5). Neither of the HTMT bias-corrected and accelerated (BCa) confidence intervals—derived from bootstrapping with 5000 samples and the no sign change option—includes the conservative threshold value of 0.85 (Franke and Sarstedt, 2019).

The bootstrapping results also show that most of the path coefficients are statistically significant (Figure 1). More specifically, ATT ($\beta = 0.419$; $SE = 0.05$) and $COMM$ ($\beta = 0.279$; $SE = 0.049$) each have significant (p value < 0.05) and positive effects on $IPTO$, whereas this is not the case for $TRUST$ ($\beta = -0.015$; $SE = 0.038$). Similarly, the $TRUST$ ($\beta = 0.251$; $SE = 0.04$) and $COMM$ ($\beta = 0.443$; $SE = 0.04$) each have significant (p value < 0.05) and positive effects on ATT . In addition, the

Table 3. Measurement model assessment.

Constructs and indicators	Mean	SD	Kurtosis	Skewness	Loading	CA	ρ_A	CR	AVE
Attitude (adapted from Ajzen and Fishbein, 1980)									
ATT1: Online travel shopping is a good idea.	5.668	1.082	−0.182	−0.595	SI	SI	SI	SI	SI
Communicability (adapted from Li and Buhalis, 2006)									
COMM1: I have heard about people booking travels online many times.	6.108	0.957	1.280	−1.137	0.850	0.828	0.829	0.897	0.744
COMM2: Many friends have purchased travels online.	5.938	1.076	0.658	−1.006	0.871				
COMM3: It is common for people to purchase travels online.	5.948	1.007	0.609	−0.910	0.867				
Intention to purchase travels online (adapted from Bigné et al., 2010; Teo and Yeong, 2003)									
IPTO1: If you were to purchase travels, the probability of purchasing online would be high.	5.690	1.130	0.261	−0.769	0.904	0.757	0.760	0.892	0.805
IPTO2: I expect to purchase travels online in the near future.	5.780	1.062	0.258	−0.718	0.890				
Trust (adapted from Kim et al., 2011)									
TRUST1: The chance of having a technical failure in an online transaction is quite small.	4.746	1.278	−0.265	−0.479	0.719	0.852	0.868	0.893	0.627
TRUST2: I believe most e-commerce travel web sites will perform to the outmost of the customers' benefit.	5.040	1.082	0.245	−0.479	0.810				
TRUST3: I believe online travel sites are trustworthy.	5.052	1.105	−0.354	−0.237	0.839				
TRUST4: Internet shopping is reliable.	5.014	1.087	−0.471	−0.187	0.814				
TRUST5: Internet shopping can be trusted and there are no uncertainties.	4.572	1.230	−0.457	−0.138	0.772				

Note: CA: Cronbach's alpha; ρ_A : construct reliability measure; CR: composite reliability; AVE: average variance extracted; SI: single item.

bootstrapping results also verify that the total effects of *COMM* ($\beta = 0.464$; $SE = 0.041$) and *TRUST* ($\beta = 0.090$; $SE = 0.042$) on the target construct *IPTO* are significant (each p value < 0.05).

Assessment of nonlinear effects

To test for potential nonlinearities in the structural model relationships, we followed Svensson et al. (2018) and conducted two tests. We first used Ramsey's (1969) RESET on the latent

Table 4. Sample demographics.

Demographic variable	Frequency	Percentage (%)
Gender		
Male	195	39.0
Female	305	61.0
Ethnicity		
Malay	292	58.4
Chinese	132	26.4
Indian	29	5.8
Others	47	9.4
Age (years)		
21–30	292	58.4
31–40	164	32.8
41–50	33	6.6
51–60	11	2.2
Level of income		
RM2001–RM3000	238	47.6
RM3001–RM4000	84	16.8
RM4001–RM5000	47	9.4
RM5001–RM6000	45	9.0
RM6001–RM7000	30	6.0
RM7001 and above	56	11.2
Do you regularly visit travel websites?		
Yes	500	100.0
No	0	0.0
Do you frequently purchase travel online?		
Yes	500	100.0
No	0	0.0

variable scores extracted after the convergence of the original model's PLS-SEM algorithm. We find that neither the partial regression of *ATT* on *TRUST* and *COMM* ($F(3,494) = 0.58, p = 0.625$) nor the partial regression of *IPTO* on *ATT*, *COMM*, and *TRUST* ($F(3,493) = 1.54, p = 0.204$) is subject to nonlinearities. Next, we included interaction terms to represent the quadratic effects between (1) *TRUST* and *COMM* on *ATT*, and (2) *TRUST*, *COMM*, and *ATT* on *IPTO*. The results of bootstrapping with 5000 samples and using no sign changes indicate that neither of the nonlinear effects is significant (Table 6). We therefore conclude that the linear effects model is robust.

Assessment of endogeneity

Our assessment of potential endogeneity follows Hult et al.'s (2018) systematic procedure, starting with the application of Park and Gupta's (2012) Gaussian copula approach, using the latent variable scores of the original model estimation as input. We first verify if the variables, which potentially exhibit endogeneity, are nonnormally distributed. We do so by running the Kolmogorov–Smirnov test with Lilliefors correction (Sarstedt and Mooi, 2019) on the latent variable scores of *ATT*, *COMM*, and *TRUST*, which serve as independent variables in the PLS path model's

Table 5. Assessment of discriminant validity using the HTMT.

	<i>ATT</i>	<i>COMM</i>	<i>IPTO</i>	<i>TRUST</i>
<i>ATT</i>				
<i>COMM</i>	0.571 [0.483; 0.646]			
<i>IPTO</i>	0.641 [0.558; 0.716]	0.620 [0.519; 0.712]		
<i>TRUST</i>	0.413 [0.324; 0.495]	0.351 [0.258; 0.444]	0.278 [0.179; 0.380]	

Note: HTMT: heterotrait-monotrait criterion; BCa: bias-corrected and accelerated. Brackets show the lower and upper bounds of the 95% BCa confidence intervals.

Table 6. Assessment of nonlinear effects.

Nonlinear relationship	Coefficient	<i>p</i> value	f^2	Ramsey's RESET
<i>TRUST*TRUST</i> → <i>ATT</i>	0.021	0.479	0.001	$F(3,494) = 0.58, p = 0.625$
<i>COMM*COMM</i> → <i>ATT</i>	0.050	0.238	0.005	
<i>TRUST*TRUST</i> → <i>IPTO</i>	0.027	0.391	0.002	$F(3,493) = 1.54, p = 0.204$
<i>ATT*ATT</i> → <i>IPTO</i>	0.012	0.789	0.000	
<i>COMM*COMM</i> → <i>IPTO</i>	0.060	0.130	0.006	

partial regressions. The results show that none of the constructs has normally distributed scores, allowing us to proceed with Park and Gupta's (2012) Gaussian copula approach. The results in Table 7 show that none of the Gaussian copulas (i.e. *ATT*, *TRUST*, and *COMM*) is significant (p value > 0.05). Specifically, considering *IPTO*'s three predictor constructs as potentially endogenous yields nonsignificant copulas of -0.010 for *ATT* (p value = 0.751), -0.042 for *TRUST* (p value = 0.816), and 0.027 for *COMM* (p value = 0.302). Note that we have also checked all other combinations of Gaussian copulas included in the model and none is significant (Table 7). We consequently conclude that endogeneity is not present in this study, which supports the robustness of the structural model results in this regard (Hult et al., 2018).

Assessment of unobserved heterogeneity

Following Sarstedt et al.'s (2017b) systematic procedure for identifying and—if required—treating unobserved heterogeneity in PLS path models, we first ran the FIMIX-PLS procedure on the data. Following Matthews et al. (2016), we initiated the procedure by assuming a one-segment solution, using the default settings for the stop criterion ($10^{-10}=1.0E-10$), the maximum number of iterations (5000), and the number of repetitions (10). To determine the maximum number of segments to extract, we first computed the minimum sample size required to estimate each segment (Sarstedt et al., 2017b). The results of a post hoc power analysis assuming an effect size of 0.15 and a power level of 80% suggest that the minimum sample size requirement is 85, which allows for extracting a maximum of five segments. We therefore reran FIMIX-PLS for two to five segments, using the same settings as in the initial analysis.

The results of the fit indices for the one- to five-segment solutions paint an ambiguous picture (Table 8). Sarstedt et al. (2011) have shown that whenever AIC_3 and CAIC indicate the same

Table 7. Assessment of endogeneity test using the Gaussian copula approach.

Test	Construct	Coefficient	p value
Gaussian copula of model 1 (endogenous variables; <i>ATT</i>)	<i>ATT</i>	0.407	0.000
	<i>TRUST</i>	−0.015	0.710
	<i>COMM</i>	0.279	0.000
	^c <i>ATT</i>	0.006	0.774
Gaussian copula of model 2 (endogenous variables; <i>TRUST</i>)	<i>ATT</i>	0.419	0.000
	<i>TRUST</i>	0.028	0.876
	<i>COMM</i>	0.279	0.001
	^c <i>TRUST</i>	−0.041	0.811
Gaussian copula of model 3 (endogenous variables; <i>COMM</i>)	<i>ATT</i>	0.417	0.000
	<i>TRUST</i>	−0.013	0.744
	<i>COMM</i>	0.235	0.000
	^c <i>COMM</i>	0.021	0.293
Gaussian copula of model 4 (endogenous variables; <i>ATT</i> , <i>TRUST</i>)	<i>ATT</i>	0.403	0.000
	<i>TRUST</i>	0.039	0.832
	<i>COMM</i>	0.279	0.000
	^c <i>ATT</i>	0.009	0.731
	^c <i>TRUST</i>	−0.051	0.769
Gaussian copula of model 5 (endogenous variables; <i>TRUST</i> , <i>COMM</i>)	<i>ATT</i>	0.418	0.000
	<i>TRUST</i>	0.041	0.820
	<i>COMM</i>	0.233	0.001
	^c <i>TRUST</i>	−0.052	0.766
	^c <i>COMM</i>	0.022	0.273
Gaussian copula of model 6 (endogenous variables; <i>ATT</i> , <i>COMM</i>)	<i>ATT</i>	0.440	0.000
	<i>TRUST</i>	−0.012	0.752
	<i>COMM</i>	0.220	0.006
	^c <i>ATT</i>	−0.011	0.698
	^c <i>COMM</i>	0.028	0.291
Gaussian copula of model 7 (endogenous variables; <i>ATT</i> , <i>TRUST</i> , <i>COMM</i>)	<i>ATT</i>	0.437	0.000
	<i>TRUST</i>	0.031	0.867
	<i>COMM</i>	0.222	0.010
	^c <i>ATT</i>	−0.010	0.751
	^c <i>TRUST</i>	−0.042	0.816
	^c <i>COMM</i>	0.027	0.302

Note: c indicates the copula term in the model.

number of segments, the results likely point to the appropriate number of segments. In our analysis, however, AIC_3 indicates a five-segment solution, whereas CAIC points to a two-segment solution. Sarstedt et al. (2011) further note that AIC_4 and Bayesian information criteria (BIC) generally perform well when used to determine the number of segments in FIMIX-PLS. Both criteria point to a two-segment solution, which appears to be densely clustered according to the EN criterion (Hair et al., 2016). A two-segment solution also meets the minimum sample size requirements for each segment (Table 9). However, minimum description length with factor 5 (MDL_5) also points to a two-segment solution. This criterion has been shown to exhibit a pronounced tendency to

Table 8. Fit indices for the one- to five-segment solutions.

Criteria	Number of segments				
	1	2	3	4	5
AIC	2424.10	2083.89	2076.62	2052.16	2044.10
AIC ₃	2431.10	2098.89	2099.62	2083.16	2083.10
AIC ₄	2438.10	2113.89	2122.62	2114.16	2122.10
BIC	2453.60	2147.11	2173.56	2182.81	2208.47
CAIC	2460.60	2162.11	2196.56	2213.81	2247.47
HQ	2435.68	2108.70	2114.66	2103.43	2108.60
MDL ₅	2627.61	2519.98	2745.30	2953.42	3177.95
LnL	-1205.05	-1026.94	-1015.31	-995.08	-983.05
EN	na	0.90	0.56	0.62	0.64
NFI	na	0.92	0.53	0.54	0.55
NEC	na	48.91	218.54	191.30	178.54

Note: AIC: Akaike's information criterion; AIC₃: modified AIC with factor 3; AIC₄: modified AIC with factor 4; BIC: Bayesian information criteria; CAIC: consistent AIC; HQ: Hannan Quinn criterion; MDL₅: minimum description length with factor 5; LnL: Log Likelihood; EN: entropy statistic; NFI: non-fuzzy index; NEC: normalized entropy criterion; na: not available; numbers in bold indicate the best outcome per segment retention criterion.

Table 9. Relative segment sizes ($N = 500$).

Number of segments	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5
1	1.000				
2	0.798	0.202			
3	0.529	0.271	0.200		
4	0.467	0.327	0.197	0.010	
5	0.426	0.389	0.091	0.050	0.044

Note: FIMIX-PLS: finite mixture partial least squares. The table shows the relative segment sizes in declining order per solution (i.e. per row); due to label switching, a specific segment can have different labels across the solutions (McLachlan and Peel, 2000); the SmartPLS 3 software uses the relative segment sizes in declining order when assigning the segment numbers to the final FIMIX-PLS segments.

underestimate the number of segments. Consequently, researchers should generally extract more segments than indicated by MDL₅ (Hair et al., 2016). Jointly, the analyses do not unambiguously point to a specific segmentation solution, because the (1) AIC₃ and CAIC point to different segment numbers, and (2) MDL₅ points to the same number of segments as AIC₄ and BIC. We therefore assume that unobserved heterogeneity is not at a critical level, which supports the results of the entire data set's analysis.³

Conclusion

Alongside the increasing dissemination of PLS-SEM in various fields of the social sciences and beyond, methodological research has produced numerous improvement to the original method (Khan et al., 2019). These advances include new estimators (Dijkstra and Henseler, 2015;

Schuberth et al., 2018), more complex modeling approaches (Becker et al., 2018; Henseler and Chin, 2010; Nitzl et al., 2016), and novel model evaluation metrics (Henseler et al., 2015; Franke and Sarstedt, 2019; Shmueli et al., 2016). A prominent stream of research in this field deals with the development of approaches to assess the robustness of the structural model results. These approaches include checking for nonlinear effects, endogeneity, and unobserved heterogeneity, which recent guidelines on PLS-SEM use consider mandatory for any analysis (Latan, 2018).

Tourism researchers have frequently considered nonlinear effects in their regression-based models, such as those between price and tourists' satisfaction (Campo and Jesús Yagüe, 2009) and global tourism competitiveness and freedom of the press (Das and DiRienzo, 2009). However, as we show in our comprehensive review of prior tourism research, the application of nonlinear effects models is scarce in the PLS-SEM context. Similarly, the application of latent class techniques such as FIMIX-PLS is scarce, not only in tourism but also in other fields of business research (Ali et al., 2018b; Cepeda Carrión and Cegarra, 2019; Ringle et al., 2019). This is surprising, considering that researchers have long called for the routine use of latent class techniques in PLS-SEM-based studies (Hair et al., 2011; Hair et al., 2012b; Latan, 2018). For example, Sarstedt and Ringle (2010: 1316) note that researchers should apply the FIMIX-PLS approach to latent class analysis to "decide whether or not unobserved heterogeneity affects path modeling outcomes." Endogeneity, on the other hand, has only recently been discussed in the context of PLS-SEM (Benitez et al., 2016; Hult et al., 2018), but research has witnessed the first applications of Gaussian copula and other approaches to identify and treat endogeneity (Benitez et al., 2018; Diallo et al., 2018; Svensson et al., 2018)—albeit outside the tourism field.

We hope that this article will increase the number of studies applying the full range of methods to test for structural model robustness. Furthermore, future research needs to extend the range of robustness tests. For example, common method variance, which refers to "variance that is attributable to the measurement method rather than to the constructs the measures are assumed to represent" (Podsakoff et al., 2003: 879), has become an increasing concern for researchers. Even though common method variance does not bias model estimates of constructs at normally reported reliability levels (Fuller et al., 2016), specific research setting may call for its control. Frequently used methods to detect and control for common method variance have, however, proven to offer limited value in the PLS-SEM context (Chin et al., 2012; Nitzl, 2016). A notable exception is Chin et al.'s (2013) measured latent marker variable approach, which has shown a promising performance in the authors' initial simulation study, but further research is needed to explore the approach's efficacy in greater detail. Further research may also extend the existing knowledge of testing whether a construct represents a composite model or a factor model (Rigdon et al., 2017; Sarstedt et al., 2016a). Initial approaches to distinguish between reflective and formative measurement models, such as the confirmatory tetrad analysis (Bollen and Ting, 1993; Bollen and Ting, 2000), which has been adapted for PLS-SEM (Gudergan et al., 2008), could be improved to support researchers by means of a statistical test that will empirically validate their assumptions. Finally, an assessment of the directionality of path relationships between the constructs in a structural model would also be beneficial. Cohen et al. (1993, 1994) suggested a technical solution for this type of analysis, which research in PLS-SEM has adopted (Callaghan et al., 2007). Even though researchers have already used this approach to confirm the selected direction of the relationships in their structural model (Sattler et al., 2010), future studies should further improve this type of analysis and translate related approaches into a PLS-SEM framework (e.g. the structural model specification searches; Marcoulides, 2003). The ultimate aim is to continuously extend the toolbox of robustness analyses and validation in PLS-SEM.

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Notes

1. Note that the REBUS-PLS method has been shown to perform poorly in latent class analysis (Ringle et al., 2013, 2014).
2. Shmueli et al. (2018) use this model and a subset of the data to illustrate Shmueli et al.'s (2016) PLSpredict procedure. However, Shmueli et al.'s (2018) presentation draws on a more complex variant of the model by illustrating PLS-SEM's predictive capabilities, which favor the use of a more complex model (Sharma et al., 2019b).
3. Based on the segmentation results, one may want to use the FIMIX-PLS method to identify extreme responses. When using a high number of segments, FIMIX-PLS extracts additional micro segments. The observations assigned to these micro segments can represent extreme and implausible responses (e.g., straight lining), which have not been removed from the data set in the prior phase of basic and descriptive data analysis. After removing such observations from the data set, the FIMIX-PLS result may point to a clear-cut segmentation solution. In our example, the elimination of the observations assigned to the micro segments may lead to a clear-cut two-segment solution (as indicated by the relative sample sizes in Table 9), but we have no indication that these responses represent extreme data. We therefore stick with the one-segment solution.

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